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Selective reduction of CO₂ to formate through bicarbonate reduction on metal electrodes: new insights gained from SG/TC mode of SECM†

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We discovered using SECM of the electro-reduction of CO₂ on a Au substrate in CO₂-saturated KHCO₃ solutions that (i) formate comes solely from the direct reduction of bicarbonate; and (ii) CO forms only from CO₂ reduction (under low pH conditions) and at higher applied potentials. The results point to the possibility of the selective reduction of CO₂ to the formate product.

Converting carbon dioxide into useful chemicals in a selective and efficient manner remains a major challenge in renewable and sustainable energy research.¹ Among the several approaches for carbon dioxide fixation,² its cathodic reduction is one of the more promising steps in the total process of carbon dioxide conversion.^{1,3} While the molecular level understanding of the reduction process has paved the way for the discovery of different molecular catalysts,^{4–6} the electrochemical reduction of CO₂ at metal electrodes⁷ in aqueous solutions leads to products like CO, HCOO[–], CH₄, C₂H₄ and alcohols in addition to H₂.^{8–10} Recently, Compton *et al.*¹¹ reviewed the work on the use of room temperature ionic liquids for the electro-reduction of CO₂ and in one of their communications¹² formic acid electro-synthesis was studied in acidified ethyl-methylimidazolium bis(trifluoromethylsulfonyl)imide. However, there is a need to evolve electrocatalysts that are effective in aqueous environments. Though the metals with high hydrogen overpotentials gave the highest faradaic efficiencies, only those with low hydrogen overpotentials offered higher energy efficiency. Hence, obtaining a balance of faradaic vs. energy efficiency with minimum hydrogen evolution requires better understanding of CO₂ reduction on various surfaces *vis-à-vis* product distribution. Nanoparticle variants of many metals¹³ have also been the subject of exciting recent research activity in the area of carbon dioxide conversion. Gold and silver are interesting electrocatalysts owing

to their ability to convert carbon dioxide to carbon monoxide selectively in aqueous solutions at room temperature.⁸

According to Hori and later workers, CO₂ electro-reduction on Au and Ag in 0.1 M KHCO₃ solution produce CO as the major product with a small percentage of formate ions.^{8,9} The proximity of the potentials at which both the CO₂ reduction and hydrogen evolution reaction (HER) take place, precludes the possibility of observing current–potential features in aqueous voltammetry. This is made more difficult as the presence of CO₂ moves the solution pH into the acidic region, and thus the HER becomes a significant interference. As the current–potential plots are invariably featureless, preliminary identification of the nature of the products poses a challenge. Hence, it is necessary to resort to other ways of detecting the reduction products online. In bulk electrolysis, the reduction is usually carried out at a fixed potential and the products are analyzed by titration methods,¹⁴ GC-MS,¹⁵ NMR¹⁶ and differential electrochemical mass spectrometry.¹⁷ However, simpler and faster methods of preliminary screening of catalytic surfaces prior to undertaking bulk electrolysis are preferred. It is advantageous that formate/formic acid, alcohols and CO, invariably found in the analysis of CO₂ reduction products are electroactive and hence amenable to electrochemical detection. Earlier work on the use of a rotating ring-disk electrode set-up for analyzing the products formed an interesting part of the electrochemical studies on CO₂ reduction.¹⁸ In comparing SECM and RRDE,¹⁹ the former is preferred as its collection efficiency is higher than that of RRDEs (improving the precision of the measurement) and is more convenient, since it allows variation of the substrate–tip distance (*d*) (compared to the fixed ring-disk spacing) and simple replacement of the substrate (compared to replacement of the disk material). In this work, we employ the substrate-generation/tip-collection (SG-TC) mode of scanning electrochemical microscopy (SECM) to sense the products of the reduction of CO₂/HCO₃[–] at a Au substrate (and Ag) using a platinum ultramicroelectrode (tip). In the SG-TC experiments the tip travels within a thick diffusion layer produced by the large substrate.¹⁹

In the course of our investigations on CO₂ reduction at Au electrodes in 0.1 M KHCO₃ solutions, we observed featureless

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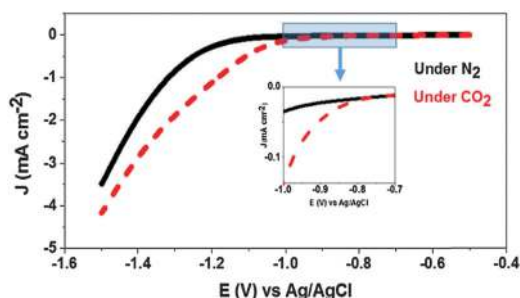


Fig. 1 Linear sweep voltammetry of CO_2 reduction on the Au substrate in 0.1 M KHCO_3 solution at a scan rate of 0.05 V s^{-1} .

voltammograms only marked by an infinite current increase at the extreme cathodic potentials both in the absence and presence of CO_2 [Fig. 1]. However, interestingly when a Pt ultramicroelectrode (Pt-UME probe) tip is placed close to the Au substrate (cathode) in the SG-TC mode of SECM [Fig. 2c], voltammetric features typical of formic acid oxidation on Pt appeared in the tip voltammograms. In addition, voltammetric features displaying the oxidation of both formic acid and CO appear when KHCO_3 is saturated with CO_2 , to a varying extent depending on the pH of the electrolyte and the applied substrate potential. We present in this communication (a) the possibility of generation of formate by direct reduction of bicarbonate alone; and (b) highlight the fact that CO forms only under low pH conditions (subsequent to the formation of formate) taking advantage of the SG-TC mode for determining the exact potentials at which the onset of product generation (here, formate and CO) occurs.

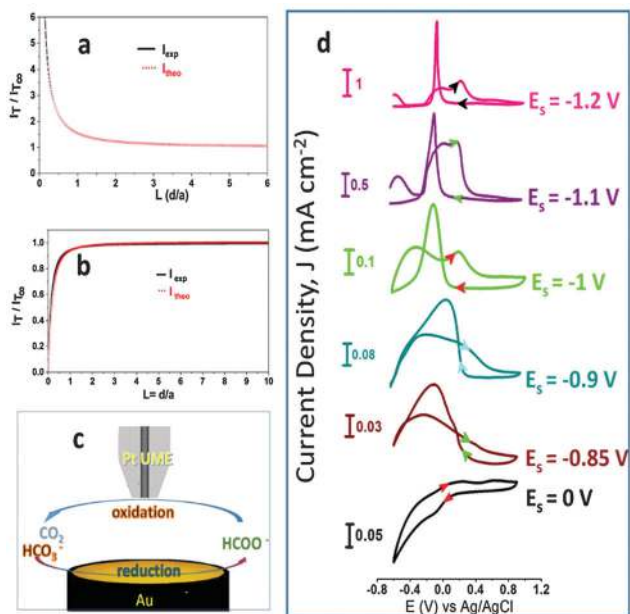


Fig. 2 (a) Positive and (b) negative approach curves of SECM using a Pt $10 \mu\text{m}$ UME; (c) schematic of the SG-TC mode for bicarbonate reduction-formate oxidation. (d) Cyclic voltammograms of the Pt UME ($10 \mu\text{m}$)-tip probe to the products (HCOO^-) generated at the Au substrate in 0.1 M KHCO_3 . Tip scan rate: 0.05 V s^{-1} .

To ensure the approach of the tip to the substrate in the SG-TC mode of SECM, a standard probe-approach curve [Fig. 2a and b] was constructed using a ferrocenemethanol probe (1 mM in 0.2 M KCl), and the probe was brought $\approx 20 \mu\text{m}$ ($d/a = 4$) (where ' a ' is the tip radius) close to the substrate (as it is appropriate to maintain $d/a \gg 1$ in order to avoid problems due to interference between diffusion layers of the substrate and the tip²⁰) before each series of experiments on bicarbonate or CO_2 reduction. A set of cyclic voltammograms at a Pt UME tip ($10 \mu\text{m}$) for the oxidation of products of the bicarbonate reduction at the Au substrate in 0.1 M KHCO_3 solution are shown in Fig. 2d. The responses were acquired at the Pt UME tip kept at a distance (d) of approx. $20 \mu\text{m}$ above the Au substrate that is biased at various cathodic potentials. As can be seen from the CV panel in Fig. 2d, at a potential of -0.85 V , features of the oxidation of small organic molecules²¹ start emerging to assume full-fledged characteristics of formic acid oxidation [Figure 1 of ref. 22b] under lower alkaline conditions (the current characteristic of the Pt UME tip in the region of -0.3 to -0.6 V (Fig. 2) is likely to arise from the oxidation of molecular hydrogen²²). ^{13}C NMR spectroscopy confirms the formation of formic acid as the main product, see the ESI,† Fig. S11.

The scenario totally changes when the bicarbonate solution is saturated with CO_2 , with the associated change in pH from 8.3 to 6.8. Depending on the substrate potential, a sharp voltammetric peak at 0.6 V vs. Ag/AgCl emerges to increase in magnitude at the expense of the response of the formic acid oxidation [Fig. 3a]. It is easy to understand the origin of this

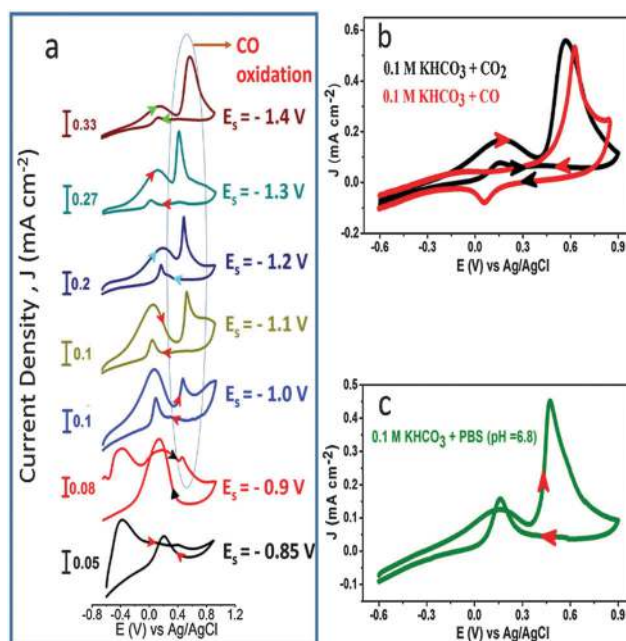


Fig. 3 (a) Cyclic voltammograms of Pt UME ($10 \mu\text{m}$)-tip probe to the products (HCOO^- and CO) generated at Au substrate in 0.1 M KHCO_3 solution saturated with CO_2 ; tip scan rate: 0.05 V s^{-1} . (b) Responses of Pt UME ($10 \mu\text{m}$)-tip probe to the oxidation of CO in CO bubbled 0.1 M KHCO_3 (red) and CO generated from Au substrate in 0.1 M KHCO_3 solution saturated with CO_2 (black); (c) 0.1 M KHCO_3 + PBS solution of pH = 6.8.

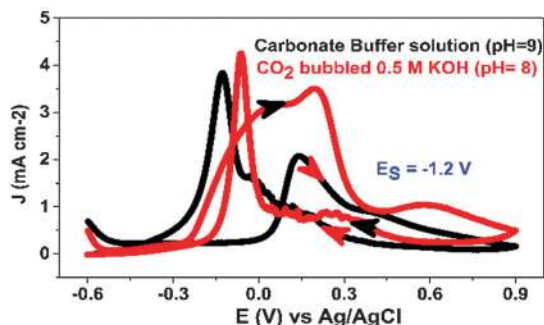
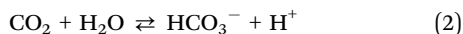
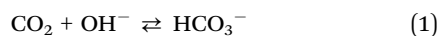


Fig. 4 Cyclic voltammetric responses of Pt UME (10 μm)-tip probe to the product (HCOO^-) generated at the Au substrate in 0.1 M carbonate buffer solution pH = 9 (black) and (ii) 0.5 M KOH solution saturated with CO_2 pH = 8 (red); tip scan rate: 0.05 V s^{-1} ; substrate potential E_S fixed at -1.2 V .

sharp feature, by recalling the conventional knowledge of CO-oxidation on a Pt surface.²³ To prove that this is due to the oxidation of CO, the following control experiments were carried out. Tip responses were acquired for the reduction of species produced at the Au substrate from (a) the bicarbonate solution acidified using a phosphate-buffer of pH 6.8 (equivalent to CO_2 -saturated bicarbonate solution) in the SG-TC mode; and (b) CO-oxidation at the Pt UME in bicarbonate solutions bubbled with carbon monoxide. The similarities in the shape and position of the anodic peak pattern in Fig. 3b and c confirm that the product of the reaction mixtures of (a) and (b) is indeed carbon monoxide, arising from CO_2 reduction on the Au surface. Whereas, the solutions of (i) $\text{CO}_2 + \text{KOH}$, and (ii) carbonate buffer (1 : 1 mol% $\text{K}_2\text{CO}_3 : \text{KHCO}_3$) yield formate as the product [Fig. 4]. That is, when electroreduction is performed in $\text{CO}_2 + 0.5 \text{ M KOH}$ solutions (pH = 8.3), only formate ions are formed. However, when the pH was changed to 6.5, CO is obtained as a major product with minor quantities of formate. The following simple equilibria are of relevance to the situation that we are describing:



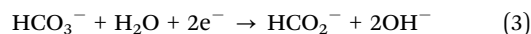
Similar results were obtained in the case of Ag with inherent differences in the position of the potentials. In spite of the reported similarities in catalytic behavior of Ag and Au, according to Hori's classification of metals with CO as the major product, the (substrate) potentials are significantly more cathodic than those observed with Au (ESI,† Fig. S1 and S2). The SG-TC mode is also unique in that it enables identification of the exact substrate potential [e.g., Fig. 2 and 3] at which the useful products are formed (formate/formic acid, and CO) with total control over the potential needed for the avoidance of hydrogen evolution.

A few control experiments involving other metal surfaces that are electrocatalytic for the reduction of (i) bicarbonate to formate; and (ii) CO_2 to formate, are required here. Direct reduction of bicarbonate to formate at a palladium electrode was established by Andre and Wrighton,²⁴ and Spichiger-Ulmann and Augustynski.²⁵ Thus, Pd can serve as a good benchmark for quantitative

comparison with our results with a Au substrate. In accordance with the results reported in ref. 25, we noticed current peaks corresponding to the direct reduction of bicarbonate in our studies [ESI,† Fig. S3 and S4] and identified that the peaks are due to the formation of formate ions, with the aid of the SG-TC mode. Based on the magnitude of the Pt UME tip currents recorded, it may be stated that the Au surface is more favorable than Pd in producing formate ions from bicarbonate.

We examined the substrate surfaces belonging to the class of metals (Hg, Bi and Sn) that are classified to yield only formic acid/formate as the product of CO_2 reduction,^{8,9} viz., Hg and Bi as well as Pd are known to produce formate from the reduction of bicarbonate.²⁶ We collected the tip responses for the products of reduction at the Hg-, Bi- and Pd-substrates. In the case of the Hg substrate, potentials as high as -2.0 V vs. Ag/AgCl were required to initiate the generation of formate from bicarbonate, while the potential is shifted 0.5 V less cathodic when the bicarbonate solution is CO_2 -saturated to produce only formate [ESI,† Fig. S5 and S6]. Bismuth-coated glassy carbon substrates also show the same trend but with reduction occurring at much lower cathodic potentials than on Hg (ESI,† Fig. S7 and S8). These results suggest that (a) formate/formic acid generation at these electrodes from CO_2 is easier than from bicarbonate; and (b) only formate/formic acid are generated until the potential limit of the hydrogen evolution reaction.

It is now clear that at Au and Ag surfaces (a) bicarbonate undergoes direct reduction; (b) in CO_2 -saturated bicarbonate solutions (pH 6.8), bicarbonate undergoes reduction first to produce formate and at higher applied potentials, CO_2 is reduced to CO; (c) in alkaline solutions (pH 8.5 to 10), CO_2 -saturated solutions yield only formate as the main product; and (d) even in slightly acidic solutions (pH 6.5), CO appears to be the major product. More importantly, it is demonstrated here that one can have control over the nature of the product species by exactly fixing the applied potential. This would suggest the possibility of obtaining selectivity between formate and CO based on the applied potentials. It will be much more advantageous if CO_2 is converted to bicarbonate which in turn undergoes reduction at lower applied potentials. The present results in juxtaposition with the suggestion of Koper *et al.*²⁶ assume significance when other more catalytic metals or alloys (for example, copper) are used. Incidentally, the local pH measurements made [ESI,† see the notes after Fig. S10] during various time intervals of electrolysis show a pH shift towards more alkalinity, pointing to a plausible reduction mechanism involving the following reactions [eqn (3) and (4)].



The present results will eventually lead to the classification of metal surfaces for bicarbonate reduction, as has been done for CO_2 reduction (*ca.* the work of Hori *et al.*). With a thesis of recent observations on the bicarbonate reduction at copper electrodes²⁶ and other studies reported on nanostructured electrocatalysts^{13,27} it will be possible to design surfaces for

selective reduction. We are currently pursuing approaches for electrocatalyst selection that in turn need deployment of product-specific ultramicroelectrode tips for *in situ* process monitoring²⁸ and for mechanistic elucidation that is more involved than the simplistic picture presented in the reactions in eqn (3) and (4).

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